1 Preface

From today(Nov 11) i will begin to re-learn quantum mechanics and start my note working ,try to fix the whole quantom world in my head.

2 Wavefunction

2.1 Begin of the probability wave

Discribe the probability of a particle that will appear in a place.And from the principle of quantom mechanics , consider a situation that don't procuducting any or annihilation any particle.

$$\int \phi^* \phi \quad \varepsilon = 1 \tag{1}$$

That means the whole probability of a partiacle is 1 for every possimble position. Here are some equations will be used in the next learning:

$$\overline{(\Delta u)^2} = \overline{(u^2)} - (\bar{u})^2 \tag{2}$$

$$\overline{\Delta u} = \sqrt{\overline{u^2} - \bar{u}^2} \tag{3}$$

These are the simplest math equations but will be used in **uncertainty relation**.

2.2 wave function

we call the wave function which is differ in a constant multipled are the same function:

$$\phi = C\phi \tag{4}$$

so we call the principle is **the probability of the function waves are relatively**.

2.3 momentum

We all know how to define the momentum of a particle , but (consider it is the relearn note) we know the uncertainty principle,so by using **Fourier transformation**:

$$f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-ikx} dx \tag{5}$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(k) e^{ikx} dk \tag{6}$$

and define the momentum:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi(x) e^{\frac{-ipx}{\hbar}} dx \tag{7}$$

What do $\phi(p)$ express?It express the **format of wave function in the space of momentum**.So we can calculate the momentum by:

$$\overline{p} = \int_p \phi(p)^* p \phi(p) \quad \epsilon \tag{8}$$

This integerallarge is defined on the space of momentum,the reason of why we not to choose to calculate the average momentum by using :$\int \phi(x)^* p \phi(x)$ is the momentum can not be certain with x at the same time.

And we also consider about the de Broglie relation:

$$p = \frac{h}{\lambda} \tag{9}$$

2.4 uncertainty relation

The strict define of uncertainty relation:

$$\Delta A \Delta B \geq \frac{1}{2} |\overline{[A, B]}| \tag{10}$$

The $\Delta A \Delta B$here are the $\sqrt{\overline{(\Delta A)^2} \times \overline{(\Delta B)^2}}$.

2.5 Average of Mechanic Variable

The define is :

$$\int \phi^* A \phi = \overline{A} \tag{11}$$

2.6 coordinate representation

How to express coordinate representation?It just like the relationship between physics object and coordinate base.Like how to get the part of a tensor or a part of a vector.The $\boldsymbol{\phi(t)}$ is the physics object and $\boldsymbol{\phi(x)}$ is this object in x space ,and $\boldsymbol{\phi(p)}$ is the object in momentum space.

And the base of these projection are the base in Hilbert space.

Hilbert space:the every possimble state of wave function ϕ.

CSCO:a set of commutating operator ,and the co-eigenstate is a base of Hlbert space.

3 Shrödinger equation

$$i\hbar \frac{\partial \phi(r,t)}{\partial t} = [-\frac{\hbar^2}{2m} \nabla + V(r,t)] \phi(r,t) \tag{12}$$

3.1 propagator

Because Shrödinger equation only have the 1 class derivative of time t so if we set the wave function **of free particle** at $t = 0$: $\phi(r,0)$ we can get we any time wave function $\phi(r,t)$:

$$\phi(r,t) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr-iEt}{\hbar}} d^3p \quad E = \frac{p^2}{2m} \tag{13}$$

$$\phi(r,0) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr}{\hbar}} d^3p \tag{14}$$

$$\text{reverse tranformation:} \phi(p) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(r,0) e^{\frac{-ipr}{\hbar}} d^3r \text{ put back to (13)} \tag{15}$$

$$\phi(r,t) = \frac{1}{2\pi\hbar^3} \int d^3p \int d^3r' \phi(r',0) e^{\frac{ip(r-r')-iEt}{\hbar}} \tag{16}$$

The reason of from eq(14) to eq(15) is **Fourier transformation don't rely on time t** and for any common situation:

$$\psi(r,t) = \frac{1}{2\pi\hbar^3} \int d^3r' \int d^3p \phi(r',t') e^{\frac{ip(r-r')-iE(t-t')}{\hbar}} \tag{17}$$

rewrite it into form :

$$\psi(r,t) = \int d^3r' G(r,r',t,t') \phi(r',t') \tag{18}$$

$$G = \frac{1}{2\pi\hbar^3} \int e^{\frac{ip(r-r')-iE(t-t')}{\hbar}} d^3p \tag{19}$$

and the G in there is called propagator.For an eigenstate of $\hat{H}$ the wave function can be write as:$\psi(r,t) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} e^{\frac{ipr-iEt}{\hbar}}$.

Now look back at eq(18) we can express this equation as :Wave function of time at t is all wave function of the particle in time t' and r' coherent superposition then spreaded to r .

3.2 Time Factor

We set $\psi(r,t) = \psi(r)f(t)$ and consider a simple situation :V in $\hat{H}$ don't explicit t.And $\psi(r,t)$ is the eigenstate of E.

$$i\hbar\psi(r)\frac{\partial f(t)}{\partial t} = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r,t) \tag{20}$$

$$E\psi(r,t) = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r,t) \tag{21}$$

$$f(t) = e^{\frac{-iEt}{\hbar}} \tag{22}$$

so the $\psi(r,t)$ can be rewirte as :$\psi_E(r)f(t)$ For a more normalized situation,we can write any $\psi(r,t)$ into eigenstate of E:$\psi_{En}(r)$:

$$\psi(r,t) = \Sigma_n C_n \psi_{En} f_n(t) = \Sigma_n C_n \psi_{En} e^{\frac{-iE_n t}{\hbar}} \tag{23}$$

$$|C_n|^2 = \int \psi^*_{En} \psi(r,0) \tag{24}$$

3.3 Stationary state

We call a wave function in form $\psi(r,t) = \psi(r)e^{-i\frac{Et}{\hbar}}$ as the stationary state wave function.When a particle stay in stationary state the energy of this particle is exactly can be determined.And the particle have these nature satisfied when it is a stationary particle:

(1) the probability density is not go with time:

$$\psi^* \psi = \rho(r) \tag{25}$$

$$\tag{26}$$

(2)the probability flux is not go with time :

$$j_n(r,t) = \frac{i\hbar}{2m}[\psi_n \nabla \psi^*_n - \psi^*_n \nabla \psi_n] = j_n(r) \tag{27}$$

(3)any average of dynamical variables which don't dependent on time will not dependent on time:

$$\overline{F} = \int \psi(r,t)^* F \psi(r,t) = \int \psi(r)^* F \psi(r) \tag{28}$$

(4)any observed value from observable dynamical variables don't explicit t will not dependent on time.

4 Operator

4.1 The Operator Algebra

Operators normally don't satisfy the law of multiple exchange.So we define the commutator:

$$[A,B] = AB - BA \tag{29}$$

And also can de defined the linearity of operators:

$$(c_1 A + c_2 B)\psi = c_1 A\psi + c_2 B\psi \tag{30}$$

4.2 commutator

Commutator is one of the important concept in QM,and there are some qualities comutator have:

$$[x, y(p(x))] = i\hbar \frac{\partial y}{\partial p(x)} \tag{31}$$

$$[p(x), y(x)] = -i\hbar \frac{\partial y}{\partial x} \tag{32}$$

The equation(31) is the equation in coordinate representation p,and equation(32) is in the coordinate representation x.So we can find the basic law:

$$[A(x), B(A(x))] = i\hbar \frac{\partial B}{\partial A(x)} \tag{33}$$

and from the upper dicussion we know there is a strong relationship between commutator and uncertainty relation at equation(10).

And the function of a variable will commmutated with another function of the same variable

$$[A(f), B(f)] = 0 \tag{34}$$

4.3 Some unique transform in QM

Most of mechanics operators is hermited ,$\hat{A}^\dagger = \hat{A}$ for define

$$\int \phi^* \hat{A}^\dagger \psi = \int (\hat{A}\psi)^* \psi = \int \psi^* A \psi \tag{35}$$

4.4 angular momentum in QM

The operator of angular momentum:$\hat{l} = r \times \hat{p} = r^i \hat{p}^j \varepsilon_{ijk}$ Rewrite them in a sphere coordinate system,noticed that the form of l_z is simple and **we know now ,the eigenstate of l_z is a Spherical harmonic function Y_{lm}.NOTICE :** the eigentstate of l_z can be a part of Sphereical harmonic function :$\psi = \frac{1}{\sqrt{2\pi}} e^{im\phi}$,and the eigenstate of l^2 can be another ppart of Spherical harmonic function.**So Spherical harmonic function is the co-eigenstate of both opeator :$\hat{l}_z, \hat{l}^2$.**

$$\hat{l}_z Y_{lm} = m\hbar Y_{lm} \tag{36}$$

$$\hat{l}^2 Y_{lm} = l(l+1)\hbar^2 Y l, m \tag{37}$$

$$[\hat{l}^2, \hat{l}_i] = 0 \tag{38}$$

$$[l_i, x^j] = i\hbar \varepsilon_{ijk} x^k \tag{39}$$

$$[\hat{l}_i, \hat{l}_j] = i\hbar \varepsilon_{ijk} \hat{l}_k \tag{40}$$

$$l_\pm Y_{lm} = \sqrt{l(l+1) + m(m \pm 1)}\hbar Y_{lm} \tag{41}$$

And the define of Shperiacl harmonic function:

$$Y_{lm}(\theta,\phi) = (-1)^m \sqrt{\frac{(l-m)!(2l+1)}{4\pi(l+m)!}} P_l^m(cos\theta)e^{im\phi} \tag{42}$$
$$l = 0, 1, 2, \ldots, m = -l, -l+1, \ldots, l-1, l.$$

And the Orthogonal normalization condition is written as:

$$\int Y_{lm}^* Y_{l'm'}\ \epsilon = \delta_{ll'}\delta_{mm'} \tag{43}$$

Now I will list a important example :
$\hat{l}^2 - \hat{l}_z^2 = \hat{l}_x^2 + \hat{l}_y^2$ and consider a situation($\psi = Y_{lm}$):$\int \psi^* \hat{l}_x^2 \psi$
$=\int \phi^* \frac{1}{i\hbar}[\hat{l}_y, \hat{l}_z]\hat{l}_x\psi = \frac{1}{i\hbar}\int \psi^*(\hat{l}_y\hat{l}_z - \hat{l}_z\hat{l}_y)\hat{l}_x\psi$
$=\frac{1}{i\hbar}\int \phi^*\hat{l}_y\hat{l}_z\hat{l}_x\phi - \phi^*\hat{l}_z\hat{l}_y\hat{l}_x\psi$
consider $\hat{l}_z\hat{l}_x = \hat{l}_x\hat{l}_z + i\hbar\hat{l}_y$
$=\frac{1}{i\hbar}\int \psi^*\hat{l}_y(\hat{l}_x\hat{l}_z + i\hbar\hat{l}_y)\psi - \psi^*\hat{l}_z\hat{l}_y\hat{l}_x\psi$
$=\frac{1}{i\hbar}\int \psi^*\hat{l}_y\hat{l}_x\hat{l}_z\psi + i\hbar\psi^*\hat{l}_y^2\psi - \psi^*\hat{l}_z\hat{l}_y\hat{l}_x\psi$
$=\frac{1}{i\hbar}\int \psi^*\hat{l}_y\hat{l}_x m\hbar\psi + i\hbar\psi^*\hat{l}_y^2\psi - m\hbar\psi^*\hat{l}_y\hat{l}_x\psi$
$=\int \psi^*\hat{l}_y^2\psi$
so we get

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} \tag{44}$$

Now consider the first equation :
$\overline{\hat{l}^2} + \overline{\hat{l}_z^2} = 2\overline{\hat{l}_y^2} = 2\overline{\hat{l}_x^2}$
and $\int \psi^*\hat{l}^2\psi = \int \psi^* l(l+1)\hbar^2\psi = l(l+1)\hbar^2$; $\int \psi^*\hat{l}_z^2\psi = \int (\hat{l}_z\psi)^*\hat{l}_z\psi = m^2\hbar^2$
so

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} = \frac{1}{2}(l(l+1) - m^2)\hbar^2 \tag{45}$$

We know the average of angular momentum $\hat{l}_x$ and $\hat{l}_y$ is : $\int Y_{lm}^*\hat{l}_x Y_{lm}\ \hat{l}_x = \int Y_{lm}^* \frac{1}{i\hbar}[\hat{l}_y, \hat{l}_z]Y_{lm} = \frac{1}{i\hbar}\int Y_{lm}^*\hat{l}_y\hat{l}_z Y_{lm} - Y_{lm}^*\hat{l}_z\hat{l}_y Y_{lm} = 0$ so we get :

$$\overline{\hat{l}_x} = \overline{\hat{l}_y} = 0 \tag{46}$$

so we know the uncertainty relation between $\hat{l}_x$ and $\hat{l}_y$

$$\Delta\hat{l}_x\Delta\hat{l}_y = \sqrt{\hat{\overline{l_x^2}} - \bar{l_x}^2}\sqrt{\hat{\overline{l_y^2}} - \bar{l_y}^2} = \frac{1}{2}(l(l+1) - m^2)\hbar \tag{47}$$

4.5 Function of operator

We can define the function of operators as :

$$F(\hat{A}) = \Sigma_n \frac{F^n(0)}{n!}\hat{A}^n \tag{48}$$

such as :$e^{a\frac{d}{dx}}$ is a typicaly function of operator.And the effect of this function is :

$$e^{a\frac{d}{dx}}\psi(x) = \Sigma_n \frac{a^n}{n!}\frac{d^n}{dx^n}\psi(x) = \psi(x+a) \tag{49}$$

so $e^{a\frac{d}{dx}}$ is the operator that move the function point from x to x+a.

5 symmetry of evolutioning Dynamical variables

5.1 Conserved observable

If a average value of observable is not changed with time , thus this observable is a conserved observable.

$$\frac{d}{dt}\bar{A}(t) = \frac{1}{i\hbar}\overline{[A,H]} \tag{50}$$

So if the operator of this observable is comuutated with Hamilton operator , this observable is a conserved observable.**Notice the detail of define described :AVERAGE VALUE is not go with time,not the classic type "value not go with time".The reason is the principle in QM is very different with classic mechanic.**And ofcourse a Conserved Observable can be a not certainll value,so the state of the operator observed can be the non-eigenstate of this operator.The key word is "average value".

5.2 Relationship Between Conserved Obserable And Energy Level Degenerate

There are three Theorems:

(A)Set two non-commutated conserverd observable F and G in a system,that means:$[F,H]=0; [G,H]=0$ but $[F,G]\neq 0$,thus the energy level usually degenerated.

(B)Inference from (A): If a conserved observable F in a system and there is a level of energy not degenerated,then ψ_E is the eigenstate of F.

(C)virial theorem:

$$2\overline{T} = \overline{\boldsymbol{r}\cdot\boldsymbol{\nabla}V} \tag{51}$$

5.3 * Schrödinger picture and Heisenberg picture

The average value derivative with t is $\frac{d}{dt}\overline{A(t)} = \frac{1}{i\hbar}\overline{[A,H]}$,now we rewrite $\psi(r,t)$ as $\hat{U}(t,0)\psi(r,0)$ and the $\hat{U}(t,0)$ here is the time shift operator and $\hat{U}(0,0) =$

1.Now we put it back into Shrödinger equation:

$$i\hbar\frac{\partial}{\partial t}\psi(r,t) = \hat{H}\psi(r,t)$$
$$i\hbar\frac{\partial}{\partial t}\hat{U}(t,0)\psi(r) = \hat{H}\hat{U}(t,0)\psi(r)$$
$$i\hbar\frac{\partial}{\partial t}\hat{U}(t,0) = \hat{H}\hat{U}(t,0)$$

so we get(if $\hat{H}$ is not go with time) :

$$\hat{U}(t,0) = e^{\frac{-i\hat{H}t}{\hbar}} \tag{52}$$

or ($\hat{H}$ go with time):

$$\hat{U}(t,0) = e^{\frac{-i\int\hat{H}dt}{\hbar}} \tag{53}$$

and we can proof:

$$\begin{aligned}\int\phi(r,t)^*\psi(r,t) &= \int\hat{U}(t,0)^*\phi(r,0)^*\hat{U}(t,0)\psi(r,0)\\ &= \int\phi(r,0)^*\phi(r,0) = 1\end{aligned} \tag{54}$$

For a more common situatioin we can say the **wave function is not go woth time but operators are go with time**,and we rewrite $\phi(r,t)$ into $\hat{U}(t,0)\psi(0)$.And the operator can be written in:

$$\begin{aligned}\overline{\hat{A}(t)} &= \int\hat{U}(t,0)^*\psi(0)^*\hat{A}\hat{U}(t,0)\psi(0)\\ &= \int\psi(0)^*\hat{U}(t,0)^\dagger\hat{A}\hat{U}(t,0)\psi(0)\\ &= \int\psi(0)^*\hat{A}(t)\psi(0)\end{aligned}$$

so we get the operator $\hat{A}(t)$ defined as

$$\hat{A}(t) = \hat{U}(t,0)^\dagger\hat{A}\hat{U}(t,0) \tag{55}$$

So the evolution equation of operators evolutino with time :

$$\begin{aligned}&\frac{d}{dt}\hat{A}(t) = \frac{d}{dt}(\hat{U}(t,0)^\dagger\hat{A}\hat{U}(t,0)) = \frac{d\hat{U}(t,0)^\dagger}{dt}\hat{A}\hat{U}(t,0) + \hat{U}(t,0)^\dagger\hat{A}\frac{d\hat{U}(t,0)}{dt}\\ &= \frac{1}{-i\hbar}\hat{U}(t,0)^\dagger\hat{H}\hat{A}\hat{(}t,0) + \frac{1}{i\hbar}\hat{U}(t,0)^\dagger\hat{A}\hat{H}\hat{U}(t,0) = \frac{1}{i\hbar}\hat{U}(t,0)^\dagger[\hat{A},\hat{H}]\hat{U}(t,0)\end{aligned}$$

Because $\hat{U}(t,0)^\dagger\hat{H}\hat{U}(t,0) = \hat{H}$(easily prooved with: A and function of A are commutated) so :

$$\frac{d}{dt}\hat{A}(t) = \frac{1}{i\hbar}[\hat{A}(t),\hat{H}] \tag{56}$$

Noticed that this is quite differernt from eq(49) this equation is not about average value but the eqaution of operator. And this kind of theory is called **Heisenberg Represent**.The eqaution : is called Heisenberg equation.As the evolution equation of operators.

5.4 The Relationship Between Conserved Observable and Symmetry

A.E.Noether:"every symmetry in nature is implicited some consverved law,the otherside every conserved law is based on one of symmetry of nature"

(A)symmetry under transformation:

$$[\hat{Q}, \hat{H}] = 0 \tag{57}$$

(B)symmetry with conserved observable:

$$[\hat{F}, \hat{H}] = 0 \tag{58}$$

(C)symmetry of space translation:

$$[p, \hat{H}] = 0 \tag{59}$$

equaly with Mommentum Conserved.
(D)symmetry of space turning translation:

$$[\vec{l}, \hat{H}] = 0 \tag{60}$$

equaly to angular momentum conserved.
(E)Some other symmetry in QM:

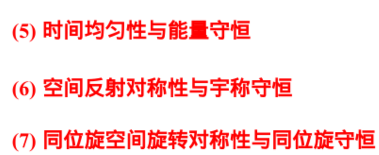

Figure 1: picture from teacher's ppt

5.5 *Commutation

Here is an example.

6 Identical Particle System and The Wavefunction

6.1 indentical particle and exchange operator

An indentical particle is the particle that have all the same intrinsic properties with other particle.Two indentical particles are indistinguishiable in QM.Define an exchange operator:

$$P_{ij}\psi(q_1 \dots q_i \dots q_j \dots) = \psi(q_1 \dots q_j \dots q_i \dots) \tag{61}$$

For any observable in a system with indentical particles ,the observables have symmetry of exchange

$$P_{ij}F(q_1 \dots q_i \dots q_j \dots) = F(q_1 \dots q_i \dots q_j \dots) \tag{62}$$
$$[p_{ij}, H] = 0 \tag{63}$$

For a system of indentical particles,we set if an exchange of indentical particles happend ,then the wavefunction of this system will only differed by times a constant.

$$P_{ij}\psi = C\psi \tag{64}$$
$$P_{ij}P_{ij}\psi = C^2\psi = \psi$$
$$C = \pm 1 \tag{65}$$

So that means only two posibility will happend :the wavefunction is symmetry:$\psi = P_{ij}\psi$.or the wavefunction is anti-symmetric:$-\psi = P_{ij}\psi$.

6.2 properties of Boson and Fermion

Boson is the particle with spin is integer times of $\hbar$.And Fermion is the particle that the spin is half-integer times of $\hbar$. But a very important nature of Boson and Fermion is Identical Boson's exchange always symmetry for a system with Boson indentical particles,and indentical Fermion's exchange in a system with indentical Fermion always anti-symmetric.

For Boson:

$$\text{if } k_1 \neq k_2$$
$$\psi^S_{k_1k_2}(q_1, q_2) = \frac{1}{\sqrt{2}}(1 + P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \tag{66}$$
$$\text{if } k_1 = k_2$$
$$\psi^S_{k_1k_2}(q_1, q_2) = \phi_{k_1}(q_1)\phi_{k_2}(q_2) \tag{67}$$

For Fermion:

$$\psi^A_{k_1k_2}(q_1, q_2) = \frac{1}{\sqrt{2}}(1 - P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \tag{68}$$

and we can see the state of Fermion at $k_1 = k_2$ is not exist.**This is the Pauli principle.**

7 Matrix

7.1 The image

Just like tensor , the image can be analog as the component,we can write the operator as the :

$$A = \hat{A}^a_b \quad ; \text{ and the component: } \quad \hat{A}^i_j = \hat{A}^a_b \phi^{*i}_a \phi^b_j \tag{69}$$

The ϕ here can be seen as the vector in GR,and the ϕ^* can be seen as the dual vector in GR.So we changed all old ways to calculate the QM questions.

7.2 relationship under a image

7.2.1 Image theory

Now we can write clearly the relationship of these operator under special image , but at first we must suppelemt a proposition:

If ϕ is the eigenstate of an operator then the operator will be a diagonal matrix:

$$\phi^{*i}\hat{F}\phi_j = \hat{F}^i_j = diag(\lambda_1, \lambda_2, \ldots, \lambda_n) \tag{70}$$

7.2.2 Eigenstate equation under a not eigenstate image

The eigenstate equation is $\hat{F}\phi = \lambda\phi$, and under the matrix way to express ,we can write it as:

$$\hat{F}^a_b \phi^b_j = \lambda_J \phi^a_j \tag{71}$$

and now we set the ψ as the base of image the we get the decompositon equation:

$$\hat{F}^a_b \phi^b_j = \lambda_J \phi^a_j = \hat{F}^i_k \psi^{*k}_b \psi^a_i \phi^b_j = \lambda_J \phi^a_j \tag{72}$$

and the eigenstate can be decomposition in new base too:

$$\phi^a_i = \phi^k_i \psi^a_k \tag{73}$$

consider all of them ,we get the final equation:

$$\hat{F}^i_k \psi^a_i \phi^k_j = \lambda_J \phi^a_j = \lambda_J \phi^k_j \psi^a_k = \lambda_J \phi^k_j \delta^i_k \psi^a_i \tag{74}$$

$$\hat{F}^i_k \phi^k_j \psi^a_i - \lambda_J \phi^k_j \delta^i_k \psi^a_i = 0 \tag{75}$$

$$(\hat{F}^i_k - \lambda_J \delta^i_k)\phi^k_j \psi^a_i = 0 \tag{76}$$

and if eq(76) have the non-normal solve of ϕ^k_j ,them the determinant of matrix $(\hat{F}^i_k - \lambda_J \delta^i_k)$ is equal to 0:

$$det(\hat{F}^i_j - \lambda_J \delta^i_j) = 0 \tag{77}$$

Which can solve the eigenstate and the eigenvalue out from base of $\hat{F}^i_j$.

7.2.3 *An example of calculation

$\psi = c_1 Y_{1-1} + c_2 Y_{11}$ express the probabily value of l_x

$\psi = c_1 Y_{1-1} + c_2 Y_{11} = c_1\psi_1 + c_2\psi_3$

$\overline{l_x} = \psi^* l_x \psi = \psi^* \dfrac{1}{2}(l_+ + l_-)\psi$

$(l_x)_{11} = \dfrac{1}{2}\psi^{*1}(l_+ + l_-)\psi_1 = 0; \quad (l_x)_{12} = \dfrac{1}{2}\psi^{*1}(l_+ + l_-)\psi_2 = \dfrac{1}{2}\sqrt{1(2)+0}\,\hbar$

$= \dfrac{\sqrt{2}}{2}\hbar; \quad (l_x)_{13} = \psi^{*1} l_x \psi_3 = \dfrac{1}{2}\psi^{*1}(l_+ + l_-)\psi_3 = 0$

$(l_x)_{22} = 0; \quad (l_x)_{23} = \dfrac{1}{2}\psi^{*2}(l_+ + l_-)\psi_3 = \dfrac{1}{2}\sqrt{2+0}\,\hbar = \dfrac{\sqrt{2}}{2}\hbar; \quad (l_x)_{33} = \dfrac{1}{2}\psi^{*3}(l_+ + l_-)\psi_3 = 0$

$l_{x,ij} = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2}\hbar & 0 \\ \frac{\sqrt{2}}{2}\hbar & 0 & \frac{\sqrt{2}}{2}\hbar \\ 0 & \frac{\sqrt{2}}{2}\hbar & 0 \end{bmatrix}; \quad \begin{vmatrix} -\lambda & \frac{\sqrt{2}}{2}\hbar & 0 \\ \frac{\sqrt{2}}{2}\hbar & -\lambda & \frac{\sqrt{2}}{2}\hbar \\ 0 & \frac{\sqrt{2}}{2}\hbar & -\lambda \end{vmatrix} = 0; \quad -\lambda(\lambda^2 - \dfrac{1}{2}\hbar^2) + \dfrac{1}{2}\lambda\hbar^2 = 0;$

$\lambda_0 = 0; \quad \lambda_1 = \hbar; \quad \lambda_{-1} = -\hbar;$

$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2}\hbar & 0 \\ \frac{\sqrt{2}}{2}\hbar & 0 & \frac{\sqrt{2}}{2}\hbar \\ 0 & \frac{\sqrt{2}}{2}\hbar & 0 \end{bmatrix}\vec{v}_0 = 0; \quad v_0^2 = 0, \quad v_0^1 = -v_0^3, \quad \vec{v}_0 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

$\begin{bmatrix} -\hbar & \frac{\sqrt{2}}{2}\hbar & 0 \\ \frac{\sqrt{2}}{2}\hbar & -\hbar & \frac{\sqrt{2}}{2}\hbar \\ 0 & \frac{\sqrt{2}}{2}\hbar & -\hbar \end{bmatrix}\vec{v}_1 = 0; \quad v_1^1 = \dfrac{\sqrt{2}}{2}v_1^2; \quad v_1^3 = \dfrac{\sqrt{2}}{2}v_1^2; \quad \vec{v}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{bmatrix}$

$\begin{bmatrix} \hbar & \frac{\sqrt{2}}{2}\hbar & 0 \\ \frac{\sqrt{2}}{2}\hbar & \hbar & \frac{\sqrt{2}}{2}\hbar \\ 0 & \frac{\sqrt{2}}{2}\hbar & \hbar \end{bmatrix}\vec{v}_{-1} = 0; \quad -v_{-1}^1 = \dfrac{\sqrt{2}}{2}v_{-1}^2; \quad -v_{-1}^3 = \dfrac{\sqrt{2}}{2}v_{-1}^2; \quad \vec{v}_{-1} = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{bmatrix}$

7.2.4 transformation between image

Just like tensor transformation :

$$\psi_i^a = \psi_i^k \phi_k^a; \psi_i^k = \psi_i^a \phi_a^{*k} \tag{78}$$

and one more thing,a special symble is also very similary with GR:

$$\psi_a^{*k} \psi_k^b = \delta_a^b \tag{79}$$

and ψ is a set of base of any image.

8 Spin and angular momentum

Spin can be seen as the new dimension of freedom.The spin operator :$\vec{S}$ and for any direct ,the projection can only be one of two values:$\frac{1}{2}\hbar$ and $-\frac{1}{2}\hbar$:

$$S_x = \pm\frac{1}{2}\hbar; S_y = \pm\frac{1}{2}\hbar; S_z = \pm\frac{1}{2}\hbar; S_n = \pm\frac{1}{2}\hbar \tag{80}$$

and the spin magnetic moment :

$$M_s = -\frac{e}{\mu}\vec{S} \tag{81}$$

8.1 the wavefunction of spin

There are two ways to write the wavefunction of spin one of them is by written into a raw:

$$\psi = a\chi_{\frac{\hbar}{2}} + b\chi_{-\frac{\hbar}{2}} \tag{82}$$

and the 'a' and 'b' here is the coefficient and the χ is eigenstate of operator S_z.That :

$$S_z\chi_{\pm\frac{\hbar}{2}} = m_s\hbar\chi_{\pm\frac{\hbar}{2}} = \pm\frac{\hbar}{2}\chi_{\pm\frac{\hbar}{2}} \quad ; m_s = \pm\frac{1}{2} \tag{83}$$

so we can write it into simple form:

$$S_z\chi_{m_s} = m_s\hbar\chi_{m_s} \; ; \; (-s < m_s < s) \tag{84}$$

$$S^2\chi_{m_s} = s(s+1)\hbar^2\chi_{m_s} \; ; \; s = \frac{1}{2} \tag{85}$$

and similar with the $\vec{l}$, $\vec{S}$ also have the relationship:

$$\begin{aligned} &[S_i, S_j] = i\hbar\varepsilon_{ijk}S_k; \\ &S_+ = S_x + iS_y; S_- = S_x - iS_y \end{aligned}$$

and the commmutator relationship:

$$[S^2, S_z] = 0; [l_i, S_j] = 0 \tag{86}$$

8.2 Pauli image

$$\sigma = \frac{2}{\hbar}s \tag{87}$$

and Pauli image in fact just the dimensionless form of s.

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \tag{88}$$

one more inportant thing

$$\sigma_+ = \frac{1}{2}(\sigma_x + i\sigma_y); \sigma_- = \frac{1}{2}(\sigma_x - i\sigma_y) \tag{89}$$

so the different between s and σ is the dimension:

$$\sigma_x = \sigma_+ + \sigma_-; \sigma_y = \frac{1}{i}(\sigma_+ - \sigma_-) \tag{90}$$

$$s_x = \frac{1}{2}(s_+ + s_-); s_y = \frac{1}{2i}(s_+ - s_-) \tag{91}$$

the coefficient is quite different here.and what's more:

$$\sigma_+\chi_- = \chi_+; \sigma_-\chi_+ = \chi_- \tag{92}$$

$$s_+\chi_- = \hbar\chi_+; s_-\chi_+ = \hbar\chi_- \tag{93}$$

8.3 angular momentum

Then we can define the angular momentum:

$$\vec{j} = \vec{s} + \vec{l} \tag{94}$$

$$\tag{95}$$

a very important thing the co-eigenstate of j^2 and j_z is :

$$\phi_{ljm_j} = C_1 Y_{lm}\chi_{\frac{1}{2}} + C_2 Y_{lm+1}\chi_{-\frac{1}{2}} \tag{96}$$

$$j_z Y_{lm+1}\chi_{-\frac{1}{2}} = (m + \frac{1}{2})\hbar Y_{lm+1}\chi_{-\frac{1}{2}} \tag{97}$$

$$j_z Y_{lm}\chi_{\frac{1}{2}} = (m + \frac{1}{2})\hbar Y_{lm}\chi_{\frac{1}{2}} \tag{98}$$

notice that the wavefunction $Y_{lm}\chi_{\frac{1}{2}}$ and $Y_{lm+1}\chi_{-\frac{1}{2}}$ have the same eigenvalue of j_z and the co-eigenstate of both (L^2, j^2, j_z) is $\phi = c_1 Y_{lm}\chi_{\frac{1}{2}} + c_2 Y_{lm+1}\chi_{-\frac{1}{2}}$ which can write into short form as : $\phi_{ljm_j}$and we can also get the simple conclusion and list all difference between three tpye of angular momentum:

$\hat{L}$	$\hat{S}$	$\hat{J}$
l	$s = \frac{1}{2}$	$j = l + \frac{1}{2} / j = l - \frac{1}{2}$
m	m_s	m_j
$-l, \ldots, l$	$-\frac{1}{2}, \frac{1}{2}$	$-j, \ldots, j$
$[L^2, L_i] = 0$	$[S^2, S_i] = 0$	$[J^2, J] = 0$
$[L_i, L_j] = i\hbar\varepsilon_{ijk}L_k$	$[S_i, S_j] = i\hbar\varepsilon_{ijk}S_k$	$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$
(L^2, L_z)	(S^2, S_z)	(L^2, J^2, J_z)
Y_{lm}	$\chi_{\frac{1}{2}}, \chi_{-\frac{1}{2}}$	ψ_{ljm_j}
$L^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}$	$S^2\chi_{m_s} = s(s+1)\hbar^2\chi_{m_s}$	$J^2\psi_{ljm_j} = j(j+2)\hbar^2\psi_{ljm_j}$
$L_z Y_{ljm_j} = m\hbar Y_{lm}$	$S_z\chi_{m_s} = m_s\hbar\chi_{m_s}$	$J_z\psi_{ljm_j} = m_j\hbar\psi_{ljm_j}$
		$L^2\psi_{ljm_j} = l(l+1)\psi_{ljm_j}$